

S/ARIMA Modelling: Session Learning Objectives

After this session, you will be able to

- 1 Identify & interpret *Moving average models*.
- 2 Examine ARMA/ ARIMA models to study suitable choice of coefficients.
- 3 Recognise how an ARIMA model may incorporate a seasonal component.
- 4 Understand Seasonality and Differencing in SARIMA models.
- 5 Interpret the ACF and PACF of various series of basic SARIMA models.
- 6 Understand the Ljung-Box residuals test to assess the model fit.
- 7 Apply lessons learned to modelling a number of datasets.

Moving Average (MA) models: Modelling the Memory of Shocks

- Moving Average models take the following form:

$$x_t = \mu + w_t + \beta_1 w_{t-1} + \cdots + \beta_q w_{t-q} = \mu + w_t + \sum_{i=1}^q \beta_i w_{t-i}$$

- μ is mean (usually zero); β_i are parameters; q is order (MA(q))
- w_t, w_{t-i} are random white noise variable parts (for 'shocks')

- 1 Can't see values of w_t , not really usual regression

-> w_t is effectively error from previous step: ($Actual_t - Predicted_t$)

- 2 MA only remembers last few w_{t-q} terms vs AR has residual lags

- 3 Patterns vs. Scale:

- > Different β_i parameters give different time series *patterns*
- > Whereas, as per AR model, variance of w_t only alters series *scale*

Moving Average Model: Ex 2.11 MA(1) & ACF

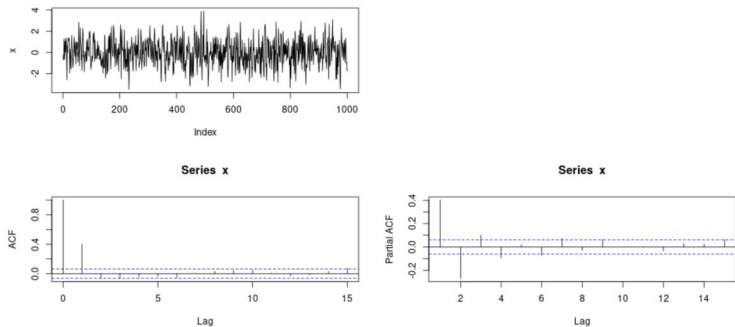


FIGURE 2.18: MA(1) & its ACF

- Top plot in Fig 2.18 shows MA(1) Series $x_t = w_t + 0.6w_{t-1}$.
- As $q = 1$, ACF has significant peak at $k = 1$ & insignificant peaks after.
- Observe the oscillating pattern in the PACF

Moving Average Model: Ex 2.12 MA(2) & ACF

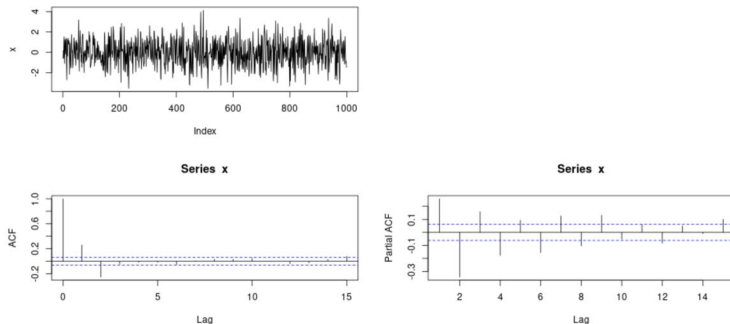


FIGURE 2.19: MA(2) & its ACF

- Top plot in Fig 2.19 shows MA(2) Series $x_t = w_t + 0.6w_{t-1} - 0.3w_{t-2}$.
- As $q = 2$, ACF has significant peak at $k = 1, 2$, insignificant peaks after.
- The oscillating pattern in the PACF is more obvious.

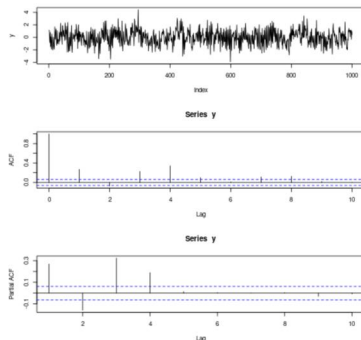
Knowing Your AR's From Your MA's

Some Useful Facts About PACF and ACF Patterns:

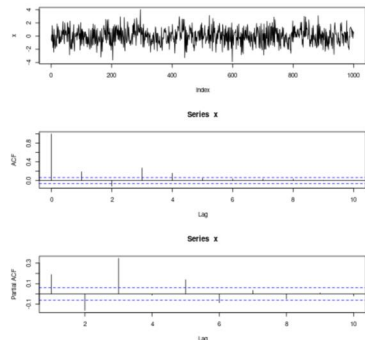
- Identification of an AR model is often best done with the PACF.
 - For AR models, theoretical PACF 'shuts off' past the model order.
 - This means, in theory, PACF autocorrelations are 0 after that.
 - Or number of non-zero partial autocorrelations gives the AR model order, p
- To identify MA model order, ACF works better than PACF.
 - MA's theoretical PACF doesn't shut off, but $\rightarrow 0$ in some manner.
 - MA model has a clearer pattern in the ACF.
 - ACF will have non-zero autocorrelations only at lags involved in the model, i.e. up to q .

Higher Order Comparisons...

- AR(4) is shown in Fig 20(a)(a): NB PACF cutoff after lag(4)
- MA(4) is shown in Fig 20(a)(b): NB ACF cutoff after lag(4)



(a) AR(4)



(b) MA(4)

FIGURE 2.20: ACF & PACF of AR(4) & MA(4) Models

A Small Change in Terminology

Types of ARMA/ ARIMA Models seen so far:

- With autoregressive terms is called an *AR model*.
- When only moving average terms involved, called *MA model*.
- With both MA & AR terms, & no differencing, is known as *ARMA*.
- Most s/w model attributes have form (AR order, differencing, MA order)
- *ARIMA* models can have AR & MA terms, and differencing operations:
 - A model with (only) two AR terms is an ARIMA of order (2,0,0)
 - A MA(2) model would be specified as an ARIMA of order (0,0,2)
 - ARIMA(1,1,1) model has 1 AR term, 1 MA term & a first differencing operation.
 - Can also have a ARIMA(1,2,1), with a second differencing operation
 - This type of difference could account for a quadratic trend in the data.

Autoregressive Moving Average (ARMA) Models: Rationale

- Looked at autoregressive & moving average processes:
 - $AR(p)$ has past behaviour as model inputs to cope with momentum & mean-reversion
 - $MA(q)$ characterises ‘shocks’, e.g. unexpected event (‘Black Swans’).
- Hence, an ARMA model attempts to capture both aspects when modelling time series³.
- $ARMA(p,q)$ is linear combination of two linear models $\therefore$ still linear: $x_t = c + \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t + \sum_{i=1}^q \beta_i w_{t-i}$
- But what's best p, q to use to model a given time series?

³NB ARMA can't handle volatility clustering & not a conditionally heteroscedastic model, so need ARCH/ GARCH.

A Summary Slide

Model	ACF	PACF
$MA(q)$: MA of order q	Cuts off after lag q	Dies down
$AR(p)$: AR of order p	Dies down	Cuts off after lag p
$ARMA(p, q)$	Dies down	Dies down
$AR(p)$ or $MA(q)$	Cuts off after lag q	Cuts off after lag p
No order AR or MA (White Noise or Random)	No Spike	No Spike

TABLE 2.2: When to Use The Different Models

Selecting Optimal ARIMA(p, d, q)

- Have seen from above and Table 2.2 that choosing optimal ARIMA(p, d, q) is inexact.
- Add to this the fact that can always approximate AR(p) by (possibly infinite) MA(q).
- Also that the order of the differencing, d , necessary to remove the trend is a judgment call.
- Thus need a more exact way to judge Optimal ARIMA(p, d, q).
- We will demonstrate this with next Example 2.13 in next slide.

Ex 2.13: Selecting Optimal (p, d, q)

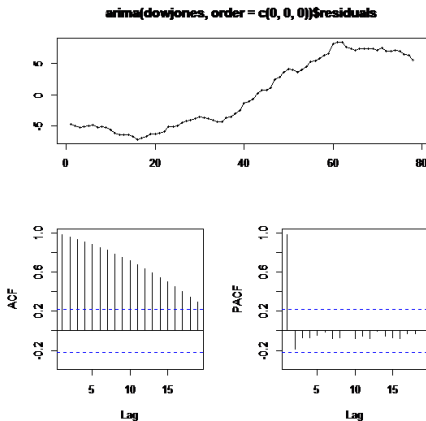


FIGURE 2.21: Residuals, ACF and PACF of Dow Jones

- In Fig 2.21, ACF has huge lags, $\Rightarrow$ trend (i.e. $y_t = y_{t-1} + \epsilon_t$) not good

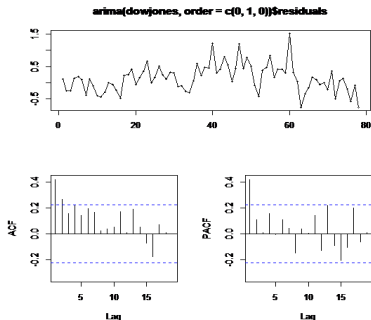
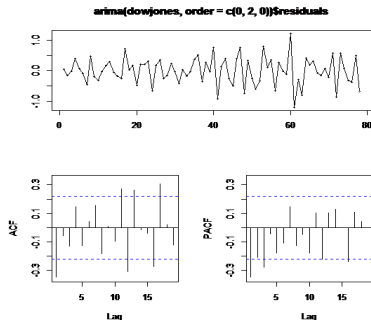
Ex 2.13: Selecting Optimal (p, d, q) (/2)(a) Dow Jones $d = 1$ (b) Dow Jones $d = 2$

FIGURE 2.22: Residuals, ACF and PACF of Differenced Dow Jones

- Looking at Fig 2.22(a), ACF still quite big values, $\Rightarrow$ still some trend in residuals
- Looking at Fig 2.22(b), trend now absent in residuals

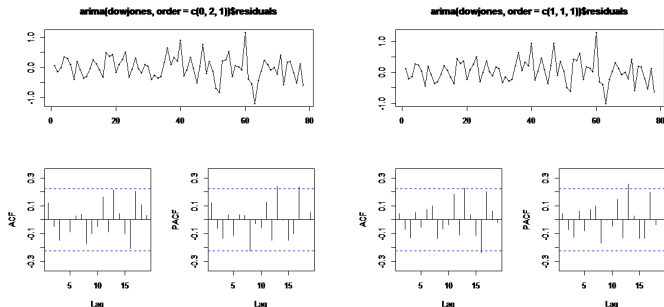
Ex 2.13: Selecting Optimal (p, d, q) (/3)(a) Dow Jones $d = 2, q = 1$ (b) Dow Jones with $(1, 1, 1)$

FIGURE 2.23: Residuals, ACF and PACF of Differenced Dow Jones

- In Fig 2.23(a) tried adding MA(1) to $d = 2$, better...
- In Fig 2.23(b), $p = 1, d = 1, q = 1$, about the same

Ex 2.13: Selecting Optimal (p, d, q) (/4)

	AIC	BIC
ARIMA(0,1,0)	95.7	98.1
ARIMA(0,2,0)	94.6	97.0
ARIMA(0,2,1)	76.4	81.1
ARIMA(1,1,1)	75.4	82.4

TABLE 2.3: AIC & BIC values of ARIMA(p, q, r) for Dow Jones Data)

- Best values of ARIMA(p, q, r) found using AIC/BIC
- Optimal ARIMA(1, 1, 1) is close to the guessed value (0, 2, 1))

A Small Refresher...

How we got to here:

- We learned earlier in the course about Time Series Components
 - For additive model, $\text{Series} = \text{Seasonal} + \text{Trend} + \text{Random}$ parts
 - The ability to fit a model to a typical series (as seen) can sometimes be problematic
 - ARIMA models work ok for most series with trend.
 - However, seasonality can still cause issues.
- A *Seasonal ARIMA* or *SARIMA model* can help here
 - Allows additional seasonal terms to ARIMA models seen above.
 - Thus, can mirror effects from a season one year to the next year

Reminder: Decomposing a Time Series

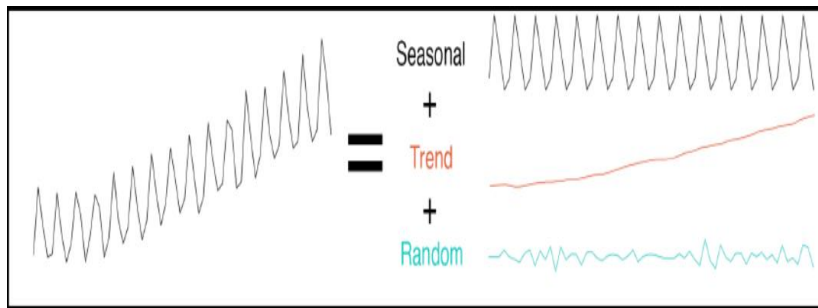


FIGURE 2.24: Time Series Decomposition

SARIMA Basics

Anatomy of a SARIMA model:

- Recall ARIMA model had the form $\text{ARIMA}(p, d, q)$
- Seasonal ARIMA model has the form $\text{ARIMA}(p, d, q)(P, D, Q)_m$
- **Red** indicate non-seasonal, and **blue**, seasonal parts of the model
 - p & seasonal P : no. of AR terms (lags of the stationarized series)
 - d & seasonal D : differencing to be done to stationarize series
 - q & seasonal Q : number of MA terms (lags of the forecast errors)
 - m : seasonal periodicity (i.e. number of time periods until pattern repeats, e.g. $m = 12$ for annual)
- As usual, must find correct values from series ACF & PACF...
 - True for basic AR/MA models, finding (P, D, Q) from ACF/ PACF is hard
 - Interaction of seasonal & non-seasonal creates issues

SARIMA Basics (/2)

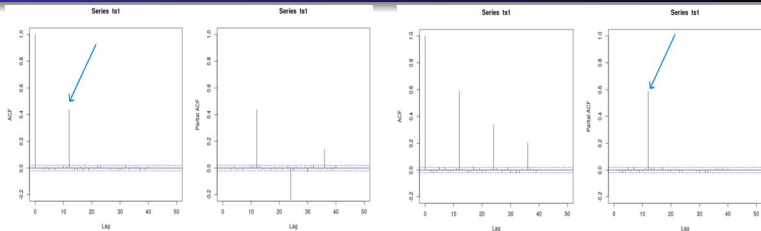
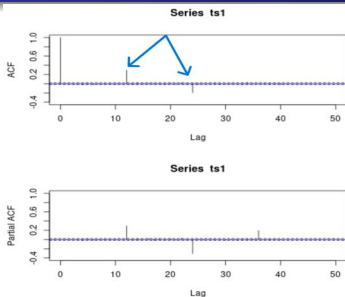
(a) $(0, 0, 0)(0, 0, 1)_{12}$ (b) $(0, 0, 0)(1, 0, 0)_{12}$

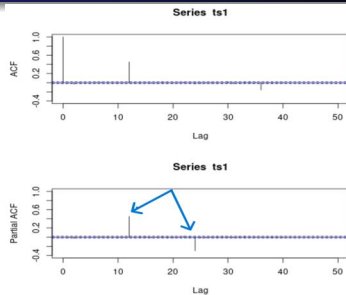
FIGURE 2.25: Seasonal ARIMA Models

- Can find seasonal part of AR or MA model in PACF, ACF seasonal lags.
- Thus, an $\text{ARIMA}(0, 0, 0)(0, 0, 1)_{12}$ model (Fig 2.25(a)) shows:
 - ACF has spike at lag 12 but no other significant spikes. → Seasonal MA(1) model
 - PACF shows exp'l decay in seasonal lags; i.e., at lags 12, 24, 36, ...
- Similarly, an $\text{ARIMA}(0, 0, 0)(1, 0, 0)_{12}$ model (Fig 2.25(b)) shows:
 - ACF has exp'l decay in seasonal lags, PACF spikes at lag 12. → Seasonal AR(1)

SARIMA Basics (/3)



(a) $(0, 0, 0)(0, 0, 2)_{12}$



(b) $(0, 0, 0)(2, 0, 0)_{12}$

FIGURE 2.26: Season & Non-Seasonal ARIMA Models

- In general, $(0, 0, 0)(0, 0, Q)_{12}$, ACF has spike(s) at $s, 2s, Qs \dots$
- > Fig 2.26(a) shows this for $\text{ARIMA}(0, 0, 0)(0, 0, 2)_{12}$ model.
- Similarly an $\text{ARIMA}(0, 0, 0)(P, 0, 0)_{12}$, PACF spikes at $s, 2s, Qs \dots$
- > 2.26(b) shows for $\text{ARIMA}(0, 0, 0)(2, 0, 0)_{12}$.

SARIMA: Identification & Best Practices

1 Model Anatomy

$$\text{SARIMA} \quad \underbrace{(p, d, q)}_{\text{Non-seasonal}} \times \underbrace{(P, D, Q)_m}_{\text{Seasonal}}$$

- (p, d, q) : Short-term dynamics.
- $(P, D, Q)_m$: Seasonal dynamics at lag m .

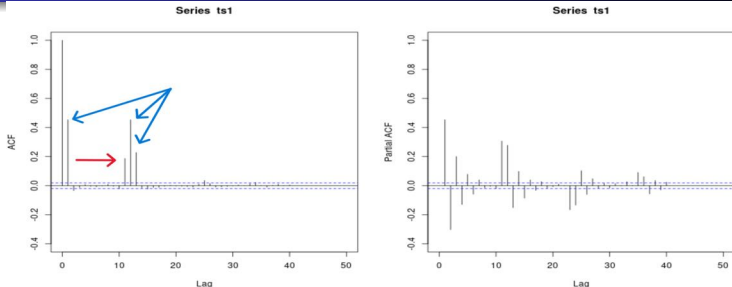
2 Identification Tips

- Seasonal Lags: Check ACF/PACF at $m, 2m, \dots$
- Parsimony: (P, D, Q) all ≤ 1 to avoid overfitting.
- Artifacts: Interactions create "artifact" spikes (e.g., $m + 1$) makes visual identifying tricky...

3 Decision Workflow

- Differencing: Do $D = 1$ before $d = 1$.
- Selection:
 - ACF/PACF is the start;
 - Use AIC/BIC to finalize.

Ex 2.14: First SARIMA models

FIGURE 2.27: $\text{ARIMA}(0, 0, 1)(0, 0, 1)_{12}$ Model

- In Fig 2.27 can see ACF/ PACF of $\text{ARIMA}(0, 0, 1)(0, 0, 1)_{12}$ model
- Simulated data, 500 values: $q = 0.7, Q = 0.6$, lag in seasons
- Lags 1, 12, 13 in ACF (blue), characteristic of $\text{ARIMA}(0, 0, 1)(0, 0, 1)_{12}$
- For subtle reasons have a non-zero AC btw x_t & $x_{t-11} \Rightarrow$ lag 11 term
- PACF tapers due to model's non-seasonal & seasonal MA terms
 - PACF tapers non-seasonally, following lag 1 and
 - PACF tapers seasonally, (near $= 12$, & again near lag $2 * S = 24$).

Ex 2.14: First SARIMA models (/2)

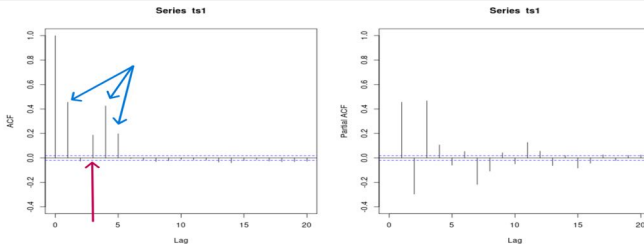


FIGURE 2.28: $\text{ARIMA}(0, 0, 1)(0, 0, 1)_4$ Model

- Fig 2.28 shows ACF/PACF of $\text{ARIMA}(0, 0, 1)(0, 0, 1)_4$ model
- Simulated data, 500 values: $q = 0.7, Q = 0.6$, lag in seasons
- As per annual model, $\text{ARIMA}(0, 0, 1)(0, 0, 1)_4$ ACF has Lags at 1, 4, 5
- For subtle reasons have a non-zero AC btw x_t & $x_{t-3} \Rightarrow$ lag 3 term
- PACF tapers due to model's non-seasonal & seasonal MA terms
 - PACF tapers non-seasonally, following lag 1 and
 - PACF tapers seasonally, (near = 4, & again near lag $2 \times S = 8$).

Ex 2.14: First SARIMA models (/3)

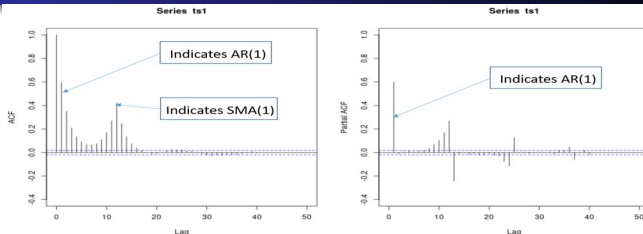


FIGURE 2.29: ARIMA(1, 0, 0)(0, 0, 1)₁₂ Model

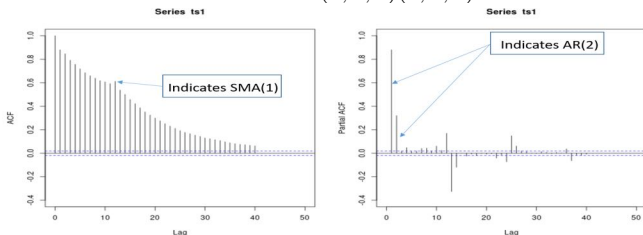


FIGURE 2.30: ARIMA(2, 0, 0)(0, 0, 1)₁₂ Model

Basics on Seasonality & Differencing

A Crucial Thing to Remember About Seasonality

- In SARIMA model, use seasonal AR & MA terms to predict x_t with data values & errors (resp) at times with lags that are multiples of S
- With monthly data, seasonal AR(1) model uses x_{t-12} to predict x_t . - Can predict this August's sales using last August's sales.
- A seasonal AR(2) uses x_{t-12} & x_{t-24} to predict x_t .
- A seasonal MA(1) model would use w_{t-12} to predict w_t .

Differencing

- Like making series trend stationary, differencing removes *Seasonal* one
- Seasonality is a regular pattern repeating every S time periods, i.e. S is the number of time periods until it repeats.
- $S = 12$ (months pa) = span of the periodic annual seasonal behaviour
- For quarterly data, $S = 4$ time periods per year.

Basics on Seasonality and Differencing (/2)

● Why Difference?

- Seasonal series are often non-stationary because mean values vary by season.
- Differencing is required to make the series mean-stationary.

● Seasonal Differencing Mechanics

- Defined as the difference from the same period last year:

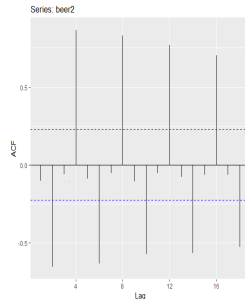
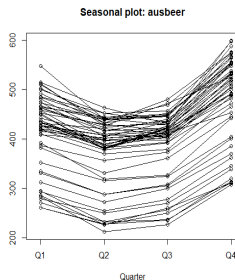
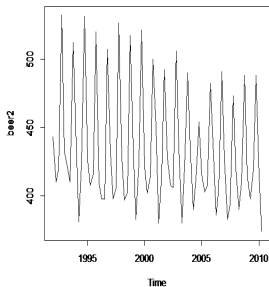
$$x_t - x_{t-S}.$$

- Monthly data ($S = 12$): $x_t - x_{t-12}$, Quarterly ($S = 4$): $x_t - x_{t-4}$.

● Comparison

- Regular differencing removes linear trends.
- Seasonal differencing removes repeating seasonal patterns.
- If differences (from last year) don't vary much for each month, Seasonal Differencing can make a series stationary

Recall a Seasonal Beer Series



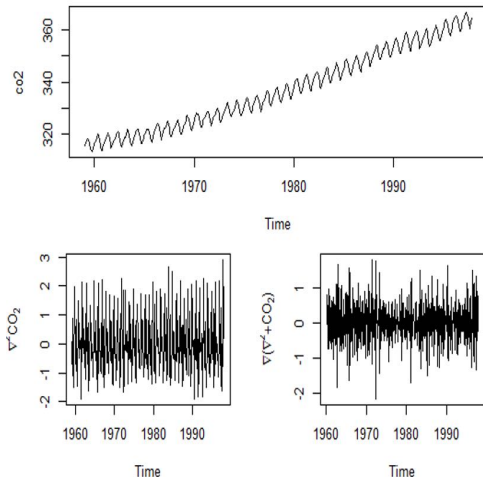
(a) Australian Beer Data (b) Seasonality per Year (c) ACF of Beer Data

FIGURE 2.31: Australian Beer Production Data from 1992 and ACF

- Australian Beer is shown in Fig 2.31(a), Seasonal in (b) and ACF in (c)
- As seen from (a) and (b) year-on-year, beer sales don't increase much
- Hence seasonal differencing would likely render the series stationary

Basics on Seasonality & Differencing (/3)

CO2 from Mauna Loa

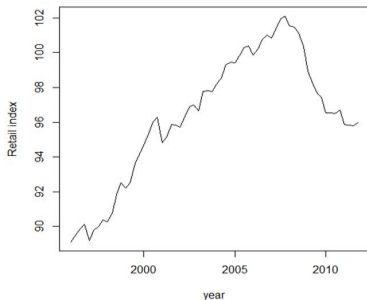


- Mauna Loa CO2 series (left) can be seasonally differenced
- Firstly, it's Double-Differenced (below, left)
- This, to remove quadratic trend.
- Then seasonally differenced (below, right) to remove remaining seasonal pattern.

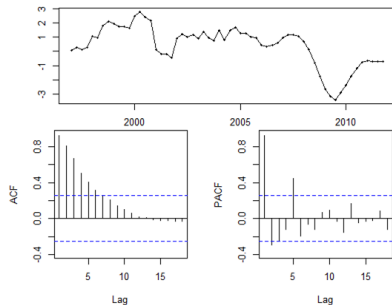
Ex 2.15: European Quarterly Retail Trade

- Fig.2.32(a) shows European Quarterly Retail Trade from 1996-2012
- And with first differences to remove non-stationarities Fig. 2.32(b)
- Still seems to be non-stationarities in the series after first differences

diff(euretail, 4)



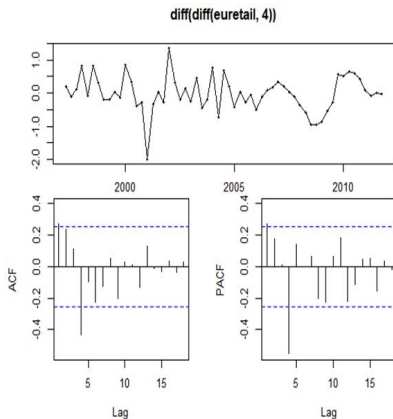
(a) Quarterly Sales



(b) P/ACF of First Diffs

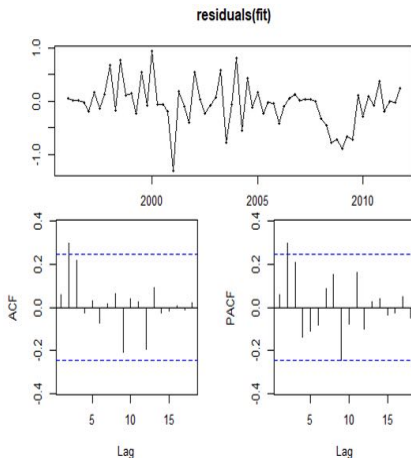
FIGURE 2.32: European Quarterly Retail Trade 1996-2012

Ex 2.15: European Quarterly Retail Trade (/2)



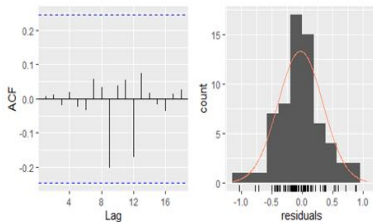
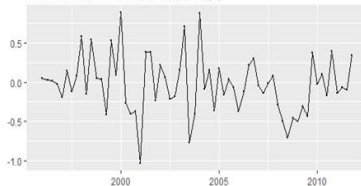
- Based on ACF & PACF shown, ACF has significant spikes
 - At lag 1 $\Rightarrow$ a non-seasonal MA(1) component.
 - Also at lag 4 $\Rightarrow$ a seasonal MA(1)
- Start with $ARIMA(0, 1, 1)(0, 1, 1)_4$ model:
 - i.e. first & seasonal difference, and
 - non seasonal and seasonal MA(1) components

Ex 2.15: European Quarterly Retail Trade (/3)



- Show residuals from an $\text{ARIMA}(0, 1, 1)(0, 1, 1)_4$ model of `euretail`
- Both ACF & PACF show spikes:
 - significant ones at lag 2&
 - almost sign't ones at lag 3,
- =>extra non seasonal terms needed in model.
- Looking at the AICc:
 - $\text{ARIMA}(0, 1, 2)(0, 1, 1)_4$ model is 74.36
 - $\text{ARIMA}(0, 1, 3)(0, 1, 1)_4$ model is 68.53
 - non seasonal & seasonal $\text{MA}(1)$ components

Ex 2.15: European Quarterly Retail Trade (/4)

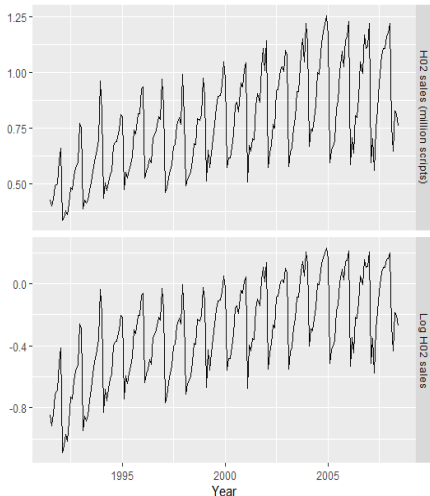
Residuals from ARIMA(0,1,3)(0,1,1)₄

- Show an $ARIMA(0, 1, 3)(0, 1, 1)_4$ model of `euretail`
- All the spikes are now within the significance limits.
- So the residuals appear to be white noise.
- Ljung-Box Test confirms no autocorrelations between residuals.

Ljung-Box Test for Autocorrelations in Residuals

- Ljung-Box test gives a different way to double check model fit.
- It is a test of autocorrelation in which it checks if any of a *group* of time series ACs differ from zero.
- The null hypothesis is that autocorrelations up to lag k equal zero (i.e. the fit is good)
- Ideally, want to *fail* to reject the null hypothesis, H_0 .
 - i.e. want to see $p > 0.05$ for the test.
 - means the residuals for our time series model are independent.
- In other words, if $p \leq 0.05 \Rightarrow$ hypothesis rejected, still remains serial correlation in the series and the model needs modification.

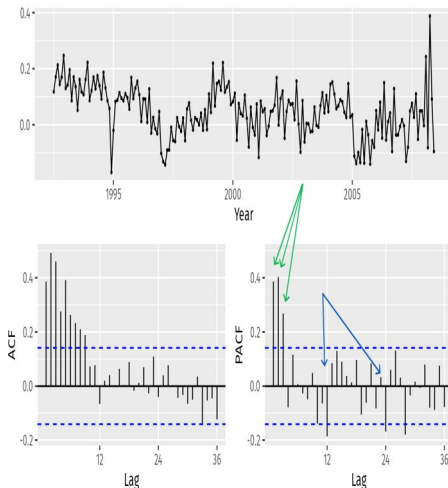
Ex 2.16: Australian Monthly Steroid Drug Subsidies



- Australian Monthly steroid drug subsidy data (1991-2008) shown (top)
- Data & figs from Hyndman
- Small increase in variance with level in top fig
- As a result take logs to stabilise variance (below fig)
- Strongly seasonal & non-stationary, so use seasonal differencing.

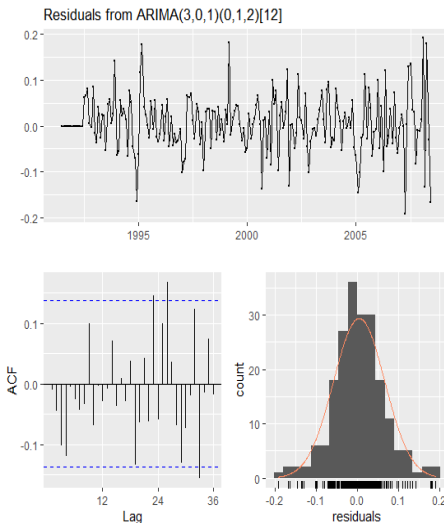
Ex 2.16: Australian Monthly Steroid Subsidies (/2)

Seasonally differenced H02 scripts



- Seasonally differenced Steroid data shown (top)
- **Seasonal**: PACF Spikes at lags 12, 24 but no ACF lags $\Rightarrow P = 2$
- **Non-seasonal**: PACF has 3 spikes, $\Rightarrow p = 3$
- > Doesn't suggest any simple model.
- > Suggests a possible $\text{ARIMA}(3, 0, 0)(2, 1, 0)_{12}$ model
- AICc modifies to $\text{ARIMA}(3, 0, 1)(0, 1, 2)_{12}$

Ex 2.16: Australian Monthly Steroid Subsidies (/3)



- Residuals shown from ARIMA(3, 0, 1)(2, 1, 0)₁₂ applied to Steroid data(top)
- Still few ACF spikes, and model fails the Ljung-Box test.
- Can still use model for forecasting,
- But prediction intervals may be inaccurate due to the correlated residuals.

S/ARIMA Modelling: Session Learning Outcomes

In this session, we covered:

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- 3 Recognising how an ARIMA model may incorporate a seasonal component.
- 4 Understanding Seasonality and Differencing in SARIMA models.
- 5 Interpreting the ACF and PACF of various series of basic SARIMA models.
- 6 Understanding the Ljung-Box residuals test to assess the model fit.
- 7 Applying lessons learned to modelling a number of datasets.